Data Management

for postdocs and researchers

MIT Libraries

What to expect

- Presentation via Powerpoint
  – Link to slides available at https://www.dropbox.com/s/ymtljsseisfs0xw/RDMresearchers_Slides_MIT.pdf?dl=0
- Ask questions anytime during the presentation.

Data Management Services @ MIT Libraries

Workshops

Upcoming Classes & Events: https://libraries.mit.edu/news/events/

Web guide

http://libraries.mit.edu/data-management

Consultations (individual or research groups)

Includes help with creating data management plans, setting up file organization, finding data repositories, etc.

Contact: data-management@mit.edu

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What we will cover

- Research as a project → Data as a project element that needs management
- Checklist for effective data management at start and end of a project
- How to check off that list: Practices, resources, and tools for practical project-based data management

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Managing Your Data – Project Start & End Checklists

This checklist (and the detailed resource pages that follow) will help you set up and maintain robust data management practices and systems through the full life of a project. For additional assistance, contact the [MIT Libraries Data Management Services](#) team at data-management@mit.edu.

When starting a project

Determine your data management needs & responsibilities

- ☐ Define what data, files, etc. you need to manage.
- ☐ Anticipate/identify funder & publisher requirements.
- ☐ Determine who on the project team is responsible for managing your data.

Store and share your work during the project

- ☐ Determine your active storage & sharing system.
- ☐ Set up a backup system.
- ☐ Develop & document/share your file organization and naming system.
- ☐ Establish access & security guidelines for your data.

Document your project

- ☐ Determine what you need to document about your data (metadata) & how to capture it.
- ☐ Document your data management system.
- ☐ Share your system with research group members, collaborators, etc.

Evaluate your system

- ☐ Test your system. Revise your system as needed over the life of the project.

Think ahead to your longer-term data management & sharing

Prepare for long-term data management in advance.

Review the "Ending a project" checklist to start establishing your practices for longer-term storage and sharing.

When ending a project

Determine post-project data sharing & restrictions

- ☐ Determine what data should be openly shared and how any confidential data will be handled.
- ☐ Make sure any funder or intellectual property restrictions on the data are documented and followed.
- ☐ Determine publisher requirements for data sharing.
- ☐ Evaluate long-term sharing and storage systems.

Store & share your work long term

- ☐ Set up long-term storage and sharing system
- ☐ Ensure those responsible for the project long term have appropriate access to your files and data.
- ☐ Make sure any needed software is accessible and properly licensed.

Document your project

- ☐ Make sure data and software README files are up to date and reflect long-term data access and storage.
- ☐ Document the published & unpublished work (manuscripts, published papers, figures) that are based on or related to your data.

Manage your records

- ☐ Store your lab notebook and other lab records according to lab protocols.

Last Updated: 2020.04.17

[Link for checklist](https://libraries.mit.edu/data-management/plan/)

https://libraries.mit.edu/data-management/plan/

Managing Your Data – Project Start & End Checklists (1)

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☐ Evaluate long-term sharing and storage systems.

Link for checklist

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Managing Your Data – Project Start & End Checklists (2)

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When starting a project

Store and share your work during the project

- ☐ Determine your active storage & sharing system.
- ☐ Set up a backup system.
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Document your project

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- ☐ Share your system with research group members, collaborators, etc.

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Document your project

- ☐ Make sure data and software README files are up to date and reflect long-term data access and storage.
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Link for checklist

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Managing Your Data – Project Start & End Checklists (4)

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When starting a project

Evaluate your system

- ☐ Test your system. Revise your system as needed over the life of the project.

Think ahead to your longer-term data management & sharing

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When ending a project

Manage your records

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[Link for checklist](#)

https://libraries.mit.edu/data-management/plan/

DETERMINING YOUR DATA MANAGEMENT NEEDS

Common data types by discipline

General	Social Sciences	STEM Fields
• images • video • mapping/GIS data • numerical measurements • software & code	• survey responses • focus group and individual interviews • economic indicators • demographics • opinion polling	• measurements generated by sensors/laboratory instruments • computer modeling • simulations • observations and/or field studies • specimen

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You'll also need to manage ...

- Analysis scripts
- Methods documentation
- Figures and graphs
- Data from outside sources
- Different versions of data as it is cleaned and processed

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Your turn! Your data!

Describe your data:

- What data products has your latest project produced? What do you anticipate generating?
- What are the different formats you use?
- What programs or code is needed to read or understand these files?

Questions funders (& you) should ask

Data description

- What is it?
- How will it be collected?
- What format is it in?
- How much data will be generated?
- Is any PII (personally identifiable information) or confidential data? Is there potentially confidential data?

Secondary data

- Are you using data that someone else produced? If so, where is it from?
- Do you have a data use agreement? Are there restrictions on accessing or sharing it?

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Funder & publisher requirements

Scholarly Publishing – MIT Libraries

Home | Open access policies at MIT | Research funder requirements | Copyright

RESEARCH FUNDER REQUIREMENTS

Major US research funder open access requirements for publications and ... below.

Many individual journals have policies requiring open access to the data that underlies a publication, for the purposes of replication and further research. Lists of such policies are available here: MIT Libraries data management site, Open Access Directory.

The Scholarly Publishing & Academic Resources Coalition (SPARC) released a website for tracking and understanding federal agencies' sharing policies for publications and data.

- White House Office of Science and Technology Policy Directive
- DOD – US Department of Defense
- DOE – US Department of Energy
- DOT – US Department of Transportation
- Environmental Protection Agency (EPA)
- Ford Foundation
- Gates Foundation

List of major funder requirements:
https://libraries.mit.edu/scholarly/research-funders/

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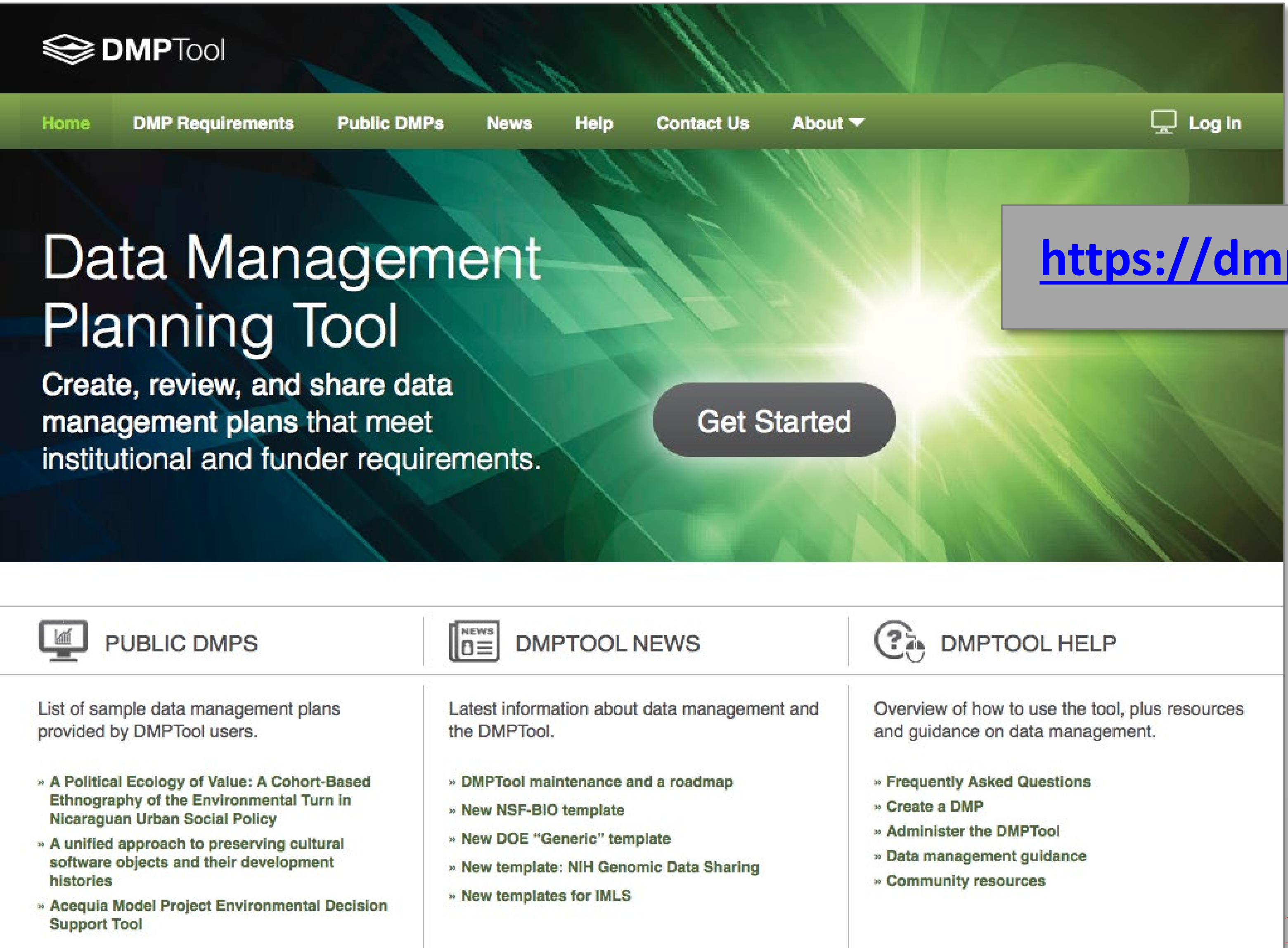
The DMPTool



https://dmptool.org

STORING & SHARING DATA

A common scenario:

- You publish a paper
- Two years later….
  - someone reads your paper and asks you for your data and code so they can replicate your work
  - You don't have access to the data in your old lab, and your analysis script was saved on your old laptop that died…

A better scenario

- You publish a paper 🎉
- You publish the data in an online repository like Zenodo. Your code in GitHub is also archived in Zenodo. You provide README files for both. ✅
- Two years later… ⏱️
  - When someone reads your paper they download the data and code from Zenodo and use the README to understand it. 👍
  - They cite your data in their paper, following the suggested citation you've given. 😃

Doing active work: back it up!

- Here + near + far backups
- Institutional resources
  - Backup services - http://ist.mit.edu/backup
  - Storage - http://ist.mit.edu/managed-servers
- Who has access to the backups?
- Who is responsible for checking them?

Here + Near + Far = 3 copies

https://commons.wikimedia.org/wiki/File%3ADesktop-PC.svg

https://pixabay.com/p-860234/ CC0

Here	Near	Far
lab computer	**portable hard drive**	**MIT's CrashPlan**
	(stored in fireproof box offsite)	(on lab computer)

Backups: Install Crashplan

https://ist.mit.edu/crashplan/

Code42 CrashPlan

Software & Hardware > CrashPlan > Code42 CrashPlan

« Back to Software Grid

CrashPlan is the recommended, free cloud backup solution for desktops and laptops at MIT. CrashPlan sends an encrypted copy of your data to Code42 Software servers via the CrashPlan server and Internet2 connection in the IS&T Data Center. TSM is the recommended service for backing up servers.

CrashPlan is available for academic and administrative use by current MIT faculty, staff, and students on both MIT-owned and personal computers.

How to Obtain

Download (MIT certificate required)

- Mac
- Windows 32-bit
- Windows 64-bit
- Linux
- Lincoln Lab: Available software (Lincoln Lab certificate required)

Installation

- Mac
- Windows
- Linux
- Upgrade instructions

GET HELP

Search the Knov
commonly aske

Request help fro

Report a security

IS&T Service De
Email: servicede
Phone: 617-253-1
Get help by ema

Note: On-site wa
this public healt

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Use open formats when possible

Proprietary Format	Alternative/Preferred Format
Excel (.xls, .xlsx)	Comma Separated Values (.csv or .tsv) ASCII
Word (.doc, .docx)	plain text (.txt), or if formatting is needed, PDF/A (.pdf)
PowerPoint (.ppt, .pptx)	PDF/A (.pdf)
Photoshop (.psd)	TIFF (.tif, .tiff)
Quicktime (.mov)	MPEG-4 (.mp4)

Not sure about a file format?
Check https://www.nationalarchives.gov.uk/PRONOM/default.htm

Library of Congress Recommended Format Specifications (for Preservation)
http://www.loc.gov/preservation/resources/rfs/

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Ethical considerations

- **Access to data**
  - Who has access to your dataset? Who needs access?
- **Private and restricted data**
  - Do you have any sensitive data?
  - Who has access to it?
- **Data source and project funder expectations**
  - Are there IP considerations in your grant/contract?

Long-term: storing data

Storage via....

- Data repositories
- With the article as supplementary material
- ~~Self-hosted storage~~

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Your turn! Your data!

Manage data for long-term access:

- What does long-term mean for your data? How long do you need to keep/store/share your data?
- Do your funder(s) or publisher(s) require any specific data-sharing mechanism?

Long term: access via repositories

Types:

- Institutional
- Discipline-specific
- General

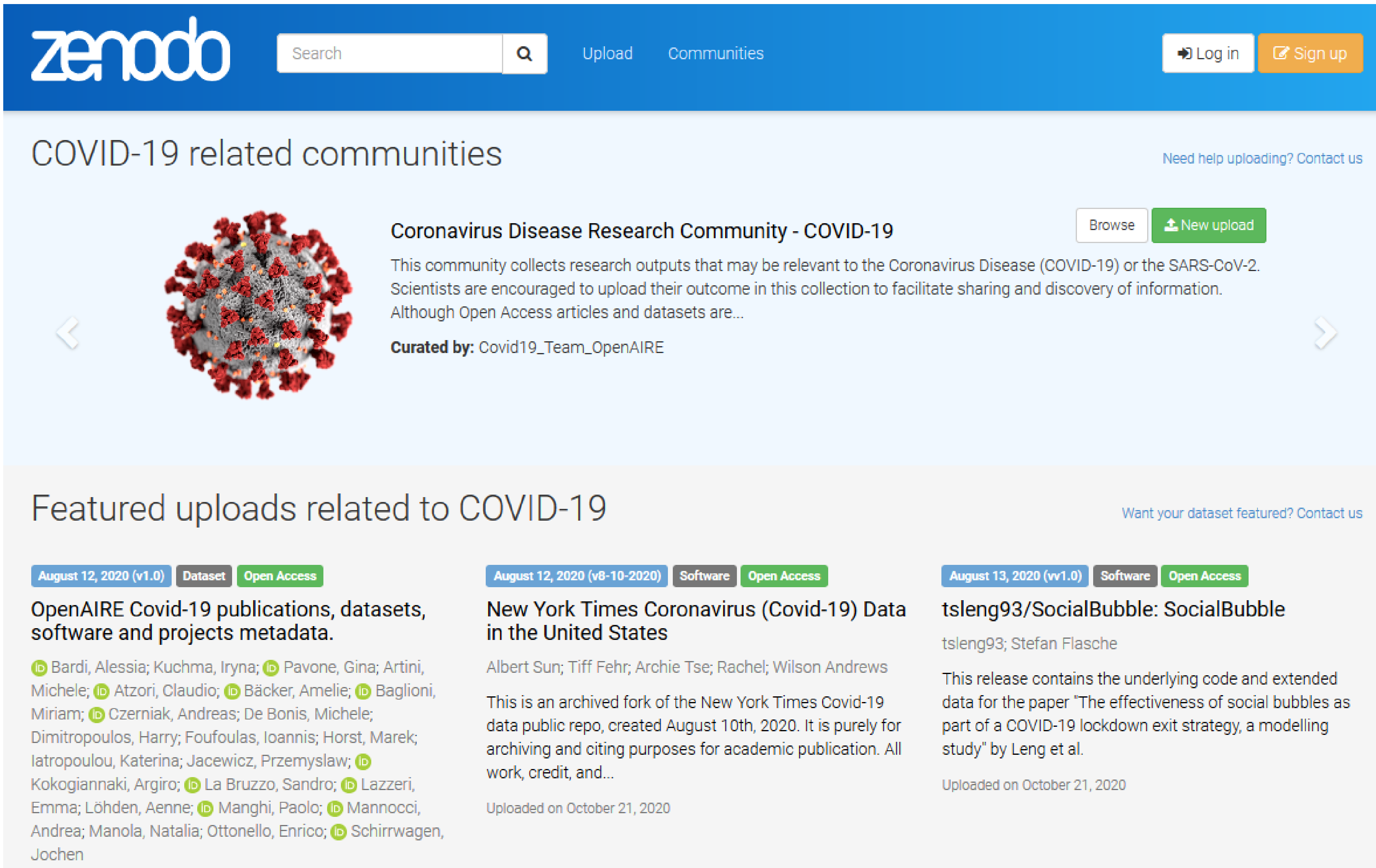
MIT's institutional repository:

dspace.mit.edu

Good for static, small datasets

http://libraries.mit.edu/data-management/share/find-repository/

Repository example: Zenodo https://zenodo.org



Connecting GitHub and Zenodo

Making Your Code Citable

10 minute read

Digital Object Identifiers (DOI) are the backbone of the academic reference and metrics system. If you're a researcher writing software, this guide will show you how to make the work you share on GitHub citable by archiving one of your GitHub repositories and assigning a DOI with the data archiving tool Zenodo.

> **ProTip:** This tutorial is aimed at researchers who want to cite GitHub repositories in academic literature. Provided you've already set up a GitHub repository, this tutorial can be completed without installing any special software. If you haven't yet created a project on GitHub, start first by uploading your work to a repository.

Choose your repository

zenodo

GitHub

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Data repositories

Things to look for:

- Open access
- Generates persistent IDs
- Offers versioning
- Good archival practices (Trusted Digital Repository certification)
- Flexible metadata
- Additional services (data cleanup, format migration/normalization, metadata assistance, etc.)

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Long term: article supplementary material

☺ +++ PROS +++ ☺

- Associates data with published articles
- Provides a citable source
- Journal requirements

☹ --- CONS --- ☹

- Limits to number and sizes of files
- Possible format limitations
- Reduced metadata
- Fragments your dataset
- May not meet funder/external requirements for your data

Data licensing

- Specifying an open license help you share your software and data openly, by telling people exactly how they can use your work
- Don’t assume something is open licensed because it’s available for free online
- https://choosealicense.com/

Discovery of data: via data papers

Data journals

- Publish "data papers"
- Help make data sets discoverable and citable
- Peer-reviewed
- Data usually stored in a repository

Data journal examples

- *Scientific Data* http://www.nature.com/sdata/about
- *Journal of Chemical and Engineering Data* http://pubs.acs.org/journal/jceaax
- *Open Health Data* http://openhealthdata.metajnl.com/
- *Earth System Science Data* http://www.earth-system-science-data.net/

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Discovery: via data citations

Why:

- Facilitates data discovery & gives researcher credit
- Recognizes data as substantial output of the research process
- Allows for citation/impact analysis, as with article publications

Include:

- Creator/author (can include **ORCID (Open Researcher and Contributor ID**)
- Title
- Publisher
- Publication date
- Version
- Persistent identifier, **e.g. a DOI (Digital Object Identifier)**

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Discovery: via persistent IDs

Common for datasets:

- DOI (digital object identifier)
- hdl (Handles)

Connecting Research and Researchers

For researchers:

- ORCID (Open Researcher and Contributor ID)

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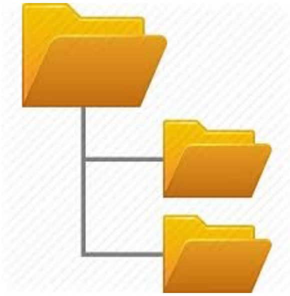
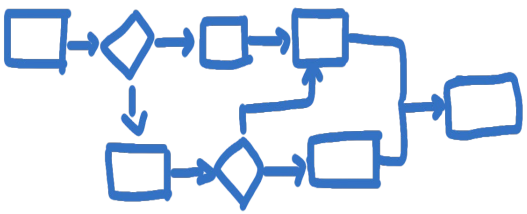
DOCUMENT YOUR PROJECT

Document! Document! Document!



Your organizational system

What are your file naming conventions? What is your folder hierarchy?



Your workflow

How did you get from raw data to the final product of research?

Your data

What is that data field? What are the units? What does that acronym mean?

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Pen exercise

What are the descriptive attributions of this object?

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What must be known to use your data?

- What is it? Why was it created?
- How to access it
- What can it be used for?
- Known problems, inconsistencies, limitations
- Collection methods, units of measure, variable names
- How to run the code. What the code does.
- Fixity checks
- Ethical/privacy restrictions
- Licensing
- Who to cite

Put it in a readMe file!

readME files

Best practices:

- Create one readME file for each data file/dataset
- Name the readME so that it's easily associated with the data file(s) it describes
- Write your readME document as a plain text file
- Format multiple readME files identically (Tip: create & use a template)
- Follow the conventions for your discipline

Resources:

- [Cornell University Readme Guides]

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Metadata: things to document

- Title…………………datasetName
- Creator……………Malinowski, Christine **with ORCID ID** http://orcid.mit.edu
- Identifier…………dataID
- Funders……………NIH
- Dates…………………..20140123-20150114
- Rights…………………..We own this data, but you can use it.
- Processing…………..Normalized
- Location……………...This file is located in this directory MyProject_NSF_2018

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readME files: Example

readme

```
title          ProjectID_20141231_cmalin.csv

description    These data were collected as part of the CMALIN Squirrel
               Avoidance Project (SAP). Data collected include squirrel
               species and presence or absence of squirrels.

format         File is a comma separated value file format, originally created
               in Microsoft Excel.

relation       ProjectID_20150115_cmalin.csv

subject        squirrels, Massachusetts, species, density

creator        Malinowski, Christine

methods        This content can go into more detail about how this data was
               created/collected.

columnHeaders  Site: Site at which the data were collected.
                              Site                    Code
                              MIT campus              MIT
                              Harvard main campus     HUM

               DateTime: Date data were collected YYYYMMDDThhmm format

funding        Collection of the data was funded by MM&I grant#123.
```

Real example: Forget, Gaël, 2016, "Documentation of https://dataverse.harvard.edu/dataverse/ECCOv4r2inputs", doi:10.7910/DVN/PICCRE, Harvard Dataverse, V2.

DMS sample: https://tinyurl.com/ybqctkpo

Inside structured data: tidy data*

- Don't leave any cells empty
- Put just one variable in a cell, don’t merge cells
- Organize the data as a single rectangle (with subjects as rows and variables as columns, and with a single header row)
- Don't use font color or highlighting as data

* Data Dictionaries

Electronic version: https://bit.ly/2JhOMUo
(make a copy or download this Google doc)

The most basic metadata

Your file names

Your file names

File naming

Naming conventions make life easier!

Consider including:

- Unique identifier (e.g., Project Name or Grant # in folder name)
- Project or research data name
- Conditions (Lab instrument, Solvent, Temperature, etc.)
- Run of experiment (sequential)
- Date (in file properties too)
- Version number or date

Be consistent:

YYYYMMDD 20201101	TimeDate DateProjectID TimeProjectID	Sample001234 Sample01234 Sample1234

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File naming: Examples

Best Practice	Example
Limit file names to 32 characters	32CharactersLooksExactlyLikeThis.csv
Don’t use special characters, . , - , or spaces	NO name&date@location.txt NO name-date–location.txt NO name.date_v1.2.txt YES name_date_location.txt
Use versioning	YES ProjID_v02.txt
Use leading zeros in sequential numbering to allow for multi-digit versions For a sequence of 1-99: 01-99 For a sequence of 1-999: 001-099-999	NO ProjID_1.csv ProjID_11.csv YES ProjID_01.csv ProjID_11.csv
Don’t use generic data file names that may conflict when moved from one location to another	NO MyData.csv YES ProjID_date.csv

Handout version downloadable from our website at:
https://libraries.mit.edu/data-management/store/organize/

File versioning

Basic: file names & changelog.txt files

- fileName_20190214
- fileName_v09
- changeLog:
  - Version name
  - What was changed?
  - Who is responsible?
  - When did it happen?
  - Why?
  - (this might also be in a readME)

Intermediate – built-in software capabilities

- Google doc history
- DropBox change tracking
- OneDrive check outs

More advanced – version control software

- Git technologies
- Concurrent versions systems (CVS)
- Apache subversion (SVN)
- Mercurial

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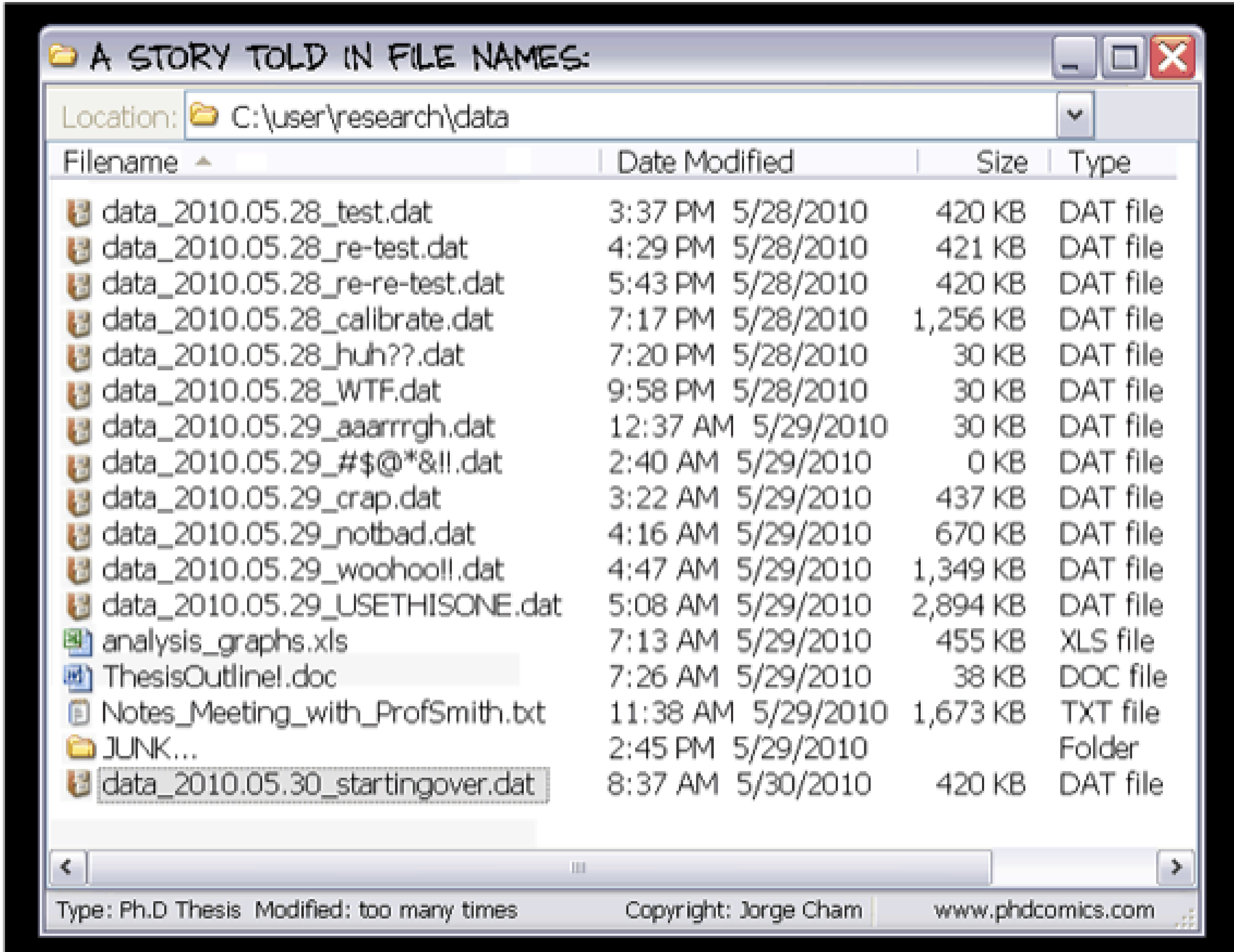
File naming & versioning

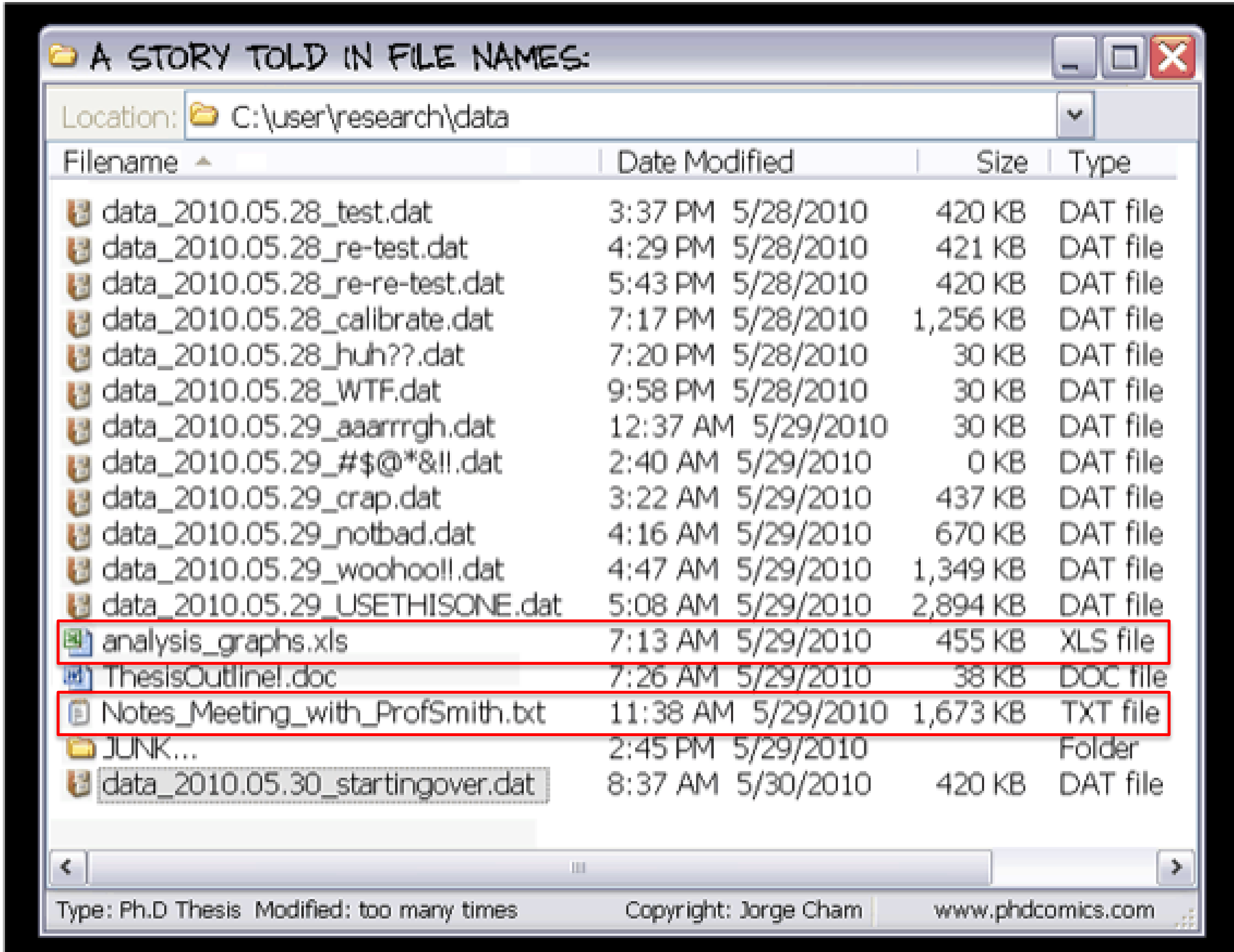
Make **a system.** ***Share*** **the system.** ***Follow*** **the system.**

Be consistent.
Be consistent.
Be consistent.

Document it.
Document it.
Document it.



http://phdcomics.com/comics/archive.php?comicid=1323



http://phdcomics.com/comics/archive.php?comicid=1323

EVALUATE YOUR SYSTEM & MANAGE YOUR RECORDS

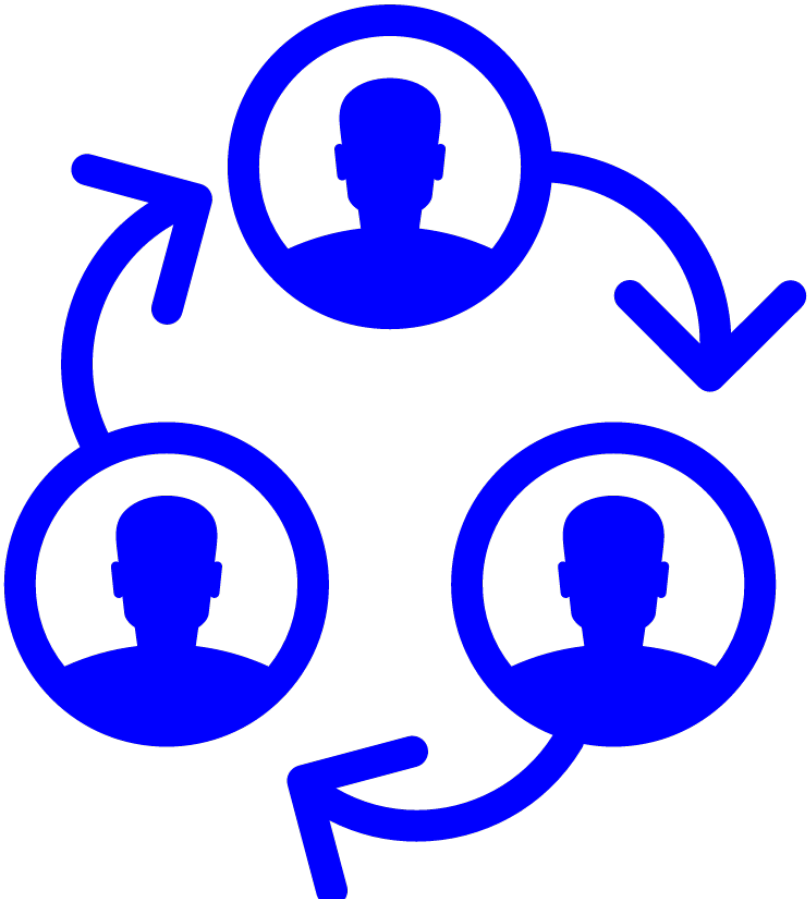
Evaluating your system

- Who is responsible?
- What are they doing?
- What tools are available?

Share your system with research group members, collaborators, etc.

Make your READMEs and other project documentation accessible to all your collaborators. Make sure everyone has a shared understanding of these documents

- Documentation & Metadata
- Cornell Library, Guide to writing "readme" style metadata
- IEEE, Creating a README File for Datasets
- DMPTool (Online tool)
- File organization README template
- Cornell Library, Guide to writing "readme" style metadata
- Organize your files
- Data Storage, backups & security

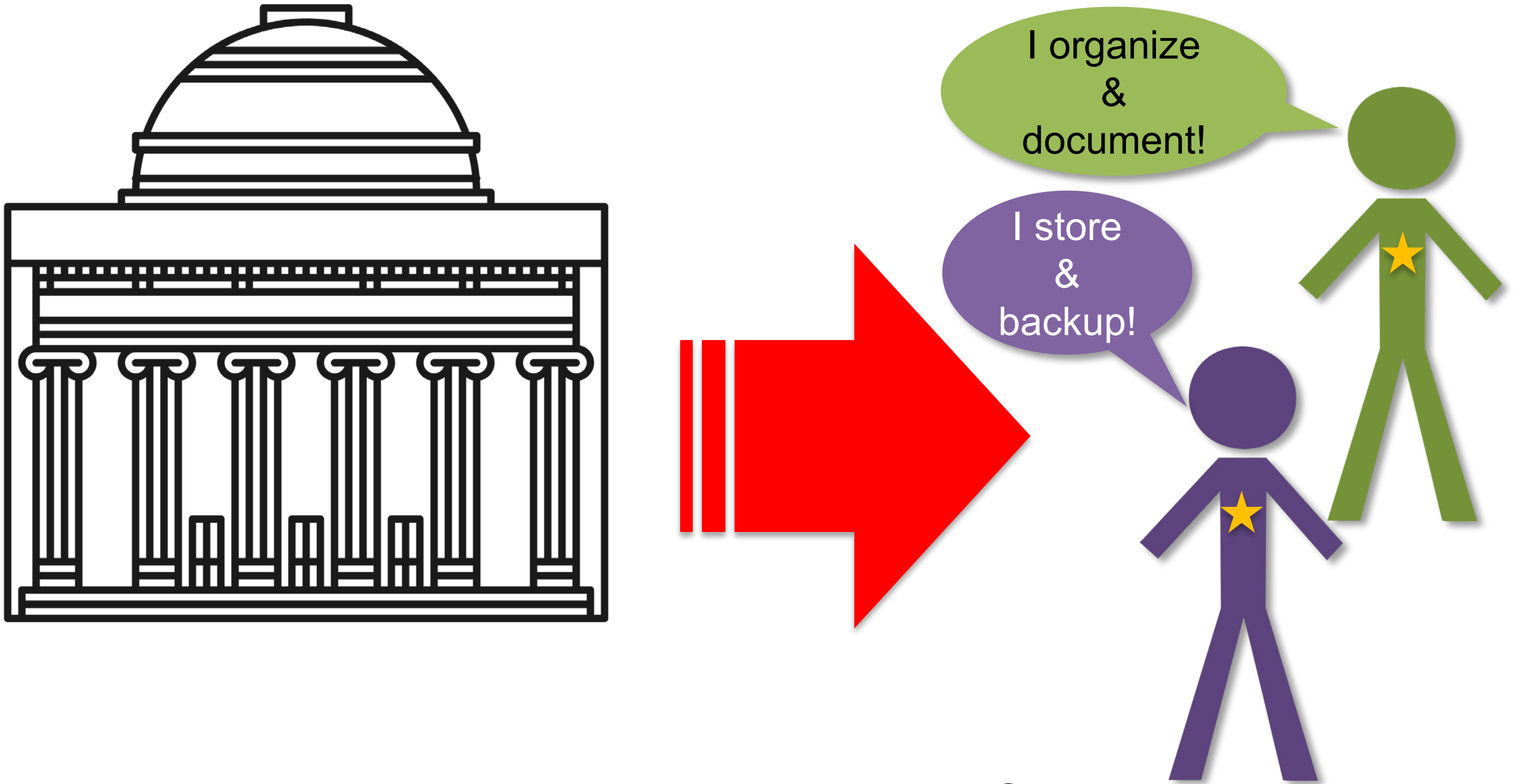


Your data collaborators

Who…

…has access? …has responsibility? …has IP rights?

What happens when…



…someone leaves?

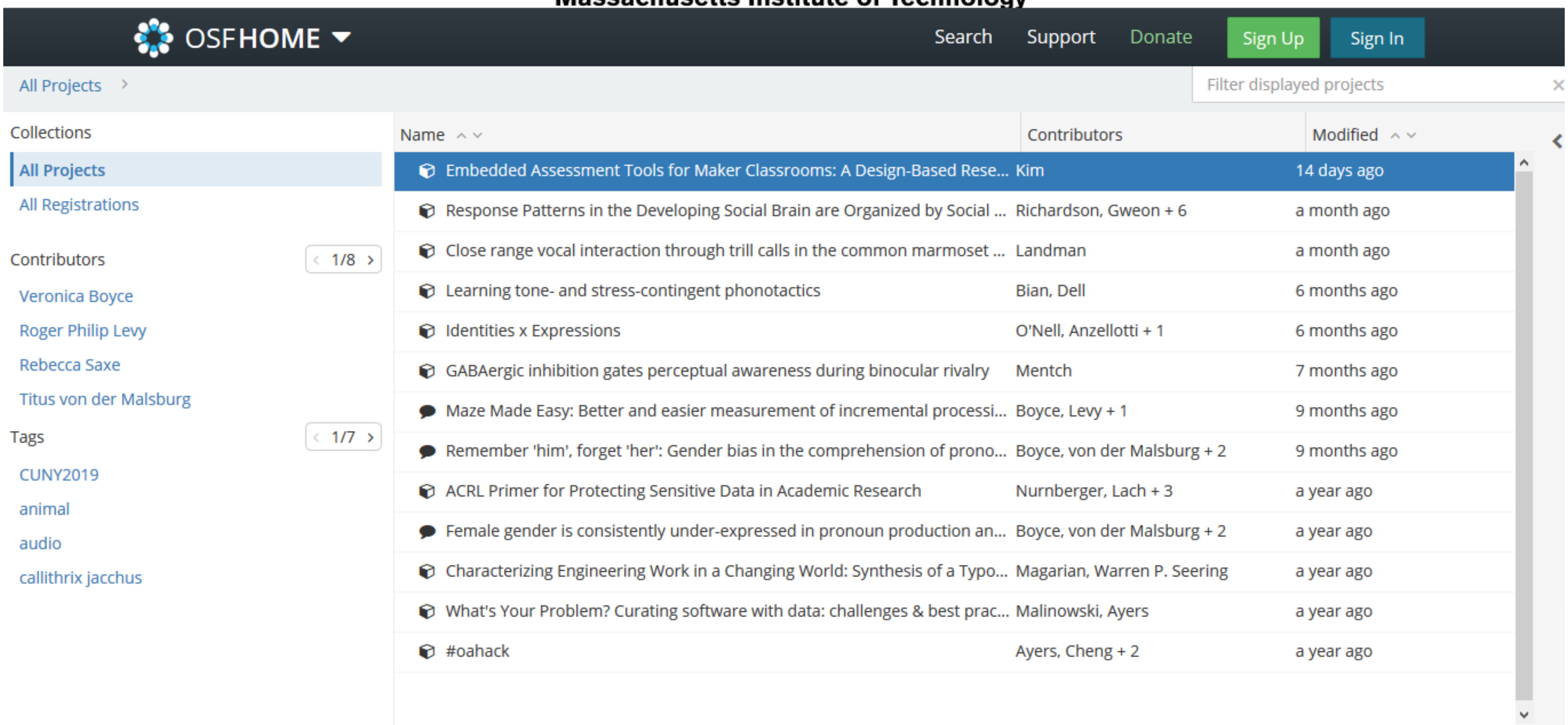
Tools for collaboration

- Dropbox for MIT
  (10TB for teaching staff, 500GB for all else)
- Open Science Framework (OSF)
  osf.io
- LabArchives
  and other electronic lab notebooks
- Collaborative writing tools
  Overleaf, Google Docs
- Versioning tools

OSF at MIT

https://osf.io/institutions/mit/

Massachusetts Institute of Technology



Name	Contributors	Modified
Embedded Assessment Tools for Maker Classrooms: A Design-Based Rese...	Kim	14 days ago
Response Patterns in the Developing Social Brain are Organized by Social ...	Richardson, Gweon + 6	a month ago
Close range vocal interaction through trill calls in the common marmoset ...	Landman	a month ago
Learning tone- and stress-contingent phonotactics	Bian, Dell	6 months ago
Identities x Expressions	O'Nell, Anzellotti + 1	6 months ago
GABAergic inhibition gates perceptual awareness during binocular rivalry	Mentch	7 months ago
Maze Made Easy: Better and easier measurement of incremental processi...	Boyce, Levy + 1	9 months ago
Remember 'him', forget 'her': Gender bias in the comprehension of prono...	Boyce, von der Malsburg + 2	9 months ago
ACRL Primer for Protecting Sensitive Data in Academic Research	Nurnberger, Lach + 3	a year ago
Female gender is consistently under-expressed in pronoun production an...	Boyce, von der Malsburg + 2	a year ago
Characterizing Engineering Work in a Changing World: Synthesis of a Typo...	Magarian, Warren P. Seering	a year ago
What's Your Problem? Curating software with data: challenges & best prac...	Malinowski, Ayers	a year ago
#oahack	Ayers, Cheng + 2	a year ago

File formats for long-term access

In the best case, your data files are both:

- Non-proprietary (also known as *open*), and
- Unencrypted

Your turn! Your data!

Define your research environment:

- Who is responsible for data management in your project/lab?
- Is there a plan for when/if they leave?
- Where/how is your data stored currently? Is it backed up? What are the plans when the grant ends?
- Does your group use collaborative tools, and if so, what works and what doesn't?

THINGS ARE STARTING TO PILE UP…

Managing Your Data – Project Start & End Checklists – again!

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Evaluate your system

- ☐ Test your system. Revise your system as needed over the life of the project.

Think ahead to your longer-term data management & sharing

Prepare for long-term data management in advance.

Review the “Ending a project” checklist to start establishing your practices for longer-term storage and sharing.

When ending a project

Determine post-project data sharing & restrictions

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- ☐ Make sure any funder or intellectual property restrictions on the data are documented and followed.
- ☐ Determine publisher requirements for data sharing.
- ☐ Evaluate long-term sharing and storage systems.

Store & share your work long term

- ☐ Set up long-term storage and sharing system
- ☐ Ensure those responsible for the project long term have appropriate access to your files and data.
- ☐ Make sure any needed software is accessible and properly licensed.

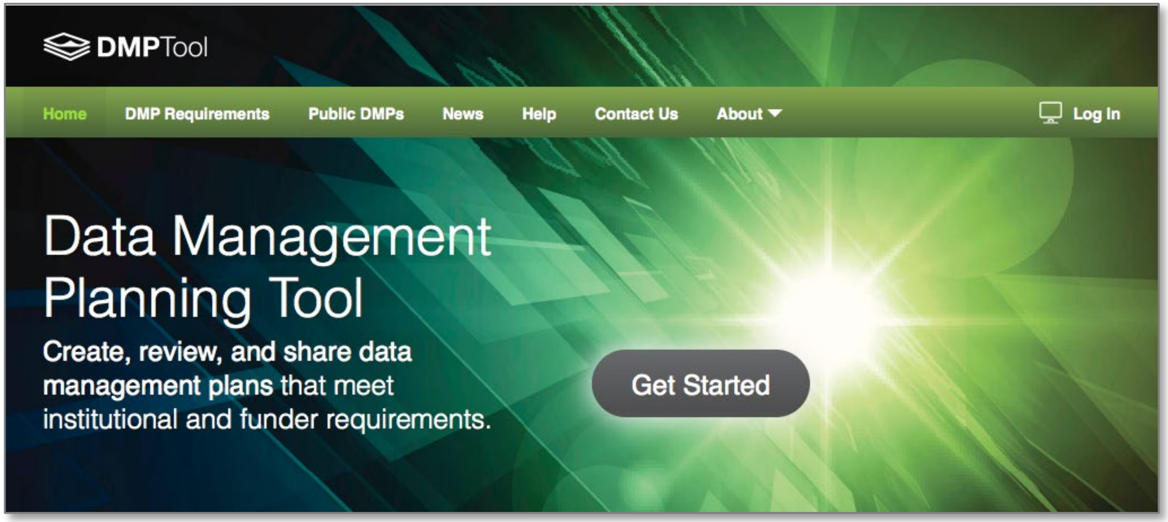
Document your project

- ☐ Make sure data and software README files are up to date and reflect long-term data access and storage.
- ☐ Document the published & unpublished work (manuscripts, published papers, figures) that are based on or related to your data.

Manage your records

- ☐ Store your lab notebook and other lab records according to lab protocols.

Last Updated: 2020-04-17



Make it a LIVING document

The DMPTool – an adaptable document!

https://dmptool.org

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NIH Final Policy Changes:

Logistics of the Final NIH Policy for Data Management and Sharing:

- Effective date: January 25, 2023
- Scope: “The DMS Policy applies to all research, funded or conducted in whole or in part by NIH, that results in the generation of scientific data.”
- Requires data management and sharing (DMS) <u>plan</u> submission **at time of application**
- Monitored compliance
- Sufficient replicability
- Data should be accessible no later than the end of the award period
- Data should be accessible via established repos as long as makes sense to community
- Funds must be spent by end of grant award period

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New definition of scientific data

Previous

"Recorded factual material commonly accepted in the scientific community as necessary to document and support research findings."

New

"The recorded factual material commonly accepted in the scientific community as of *sufficient quality to validate and replicate research findings, regardless of whether the data are used to support scholarly publications*."

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Sharing timelines

“[s]hared scientific data should be made accessible *as soon as possible*, and *no later than the time of an associated publication, or the end of the award/support period, whichever comes first*.”

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Data retention

“Researchers are encouraged to consider relevant requirements and expectations (e.g., data repository policies, award record retention requirements, journal policies) as guidance for the minimum time frame that scientific data should be made available, which researchers may extend.”

“NIH encourages researchers to **make scientific data available for as long as they anticipate it being useful** for the larger research community, institutions, and/or the broader public.”

Summary: **It depends…**

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If you have questions…

- About data: data-management@mit.edu
- About copyright: scholarlypub@mit.edu
- About any part of the research lifecycle
  https://libraries.mit.edu/scholarly/

Any library question!

http://libraries.mit.edu/ask

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Questions? Comments? Experiences to share?

Check out our web site:
http://libraries.mit.edu/data-management

Contact: data-management@mit.edu

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